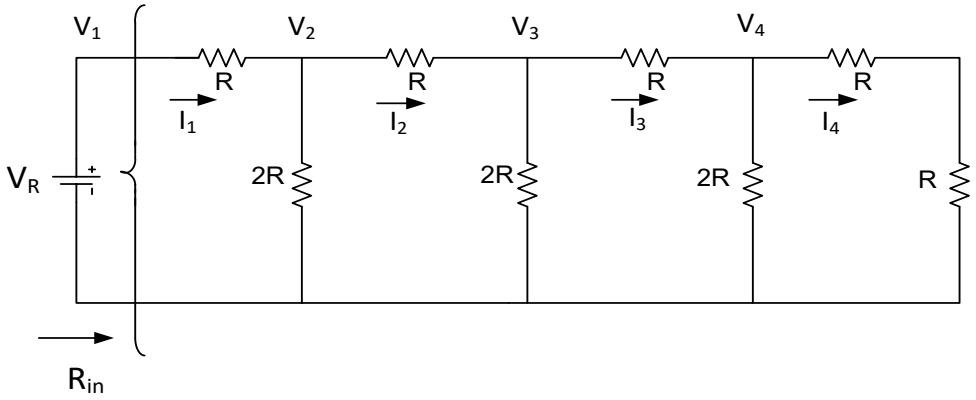
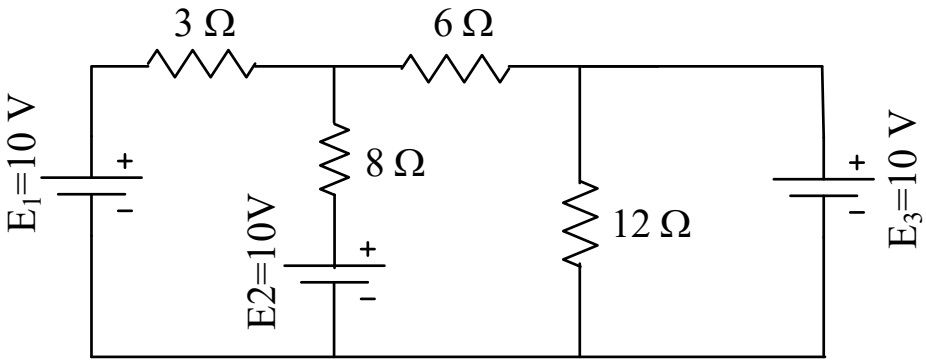
EE1100: Basic Electrical Engineering, July-Nov. 2026
Tutorial 1: Circuit Laws and Theorems

Q1. Figure below shows a R-2R network that is often used in D-A converters. Find $V_{1,}$,V2, V3 and V_4 in terms of V_R. What is the input resistance R seen at R_{in}. Also, find I_1, I_2, I_3 and I_4.



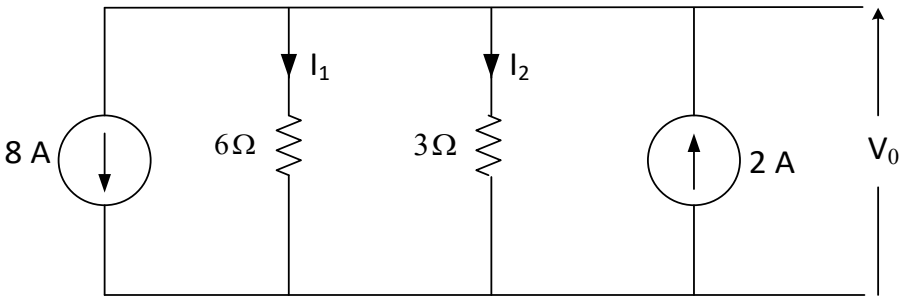
[Ans.: V_R, $V_R/2$, $V_R/4$, $V_R/8$, R_{in}=2 R, $V_R/(2*R)$, $V_R/(4*R)$, $V_R/(8*R)$, $V_R/(16*R)$]

Q2. By applying Kirchhoff's voltage law, determine the current which flows through each resistor in the circuit, shown below.



[Ans.: $I_{3\Omega} = 0$, $I_{8\Omega} = 0$, $I_{6\Omega} = 0$, $I_{12\Omega} = 0.833$ A (from top to bottom)]

Q3. Find V_0, $I_{1,}$ I_2, $P_{6\Omega}$ in the network shown below.

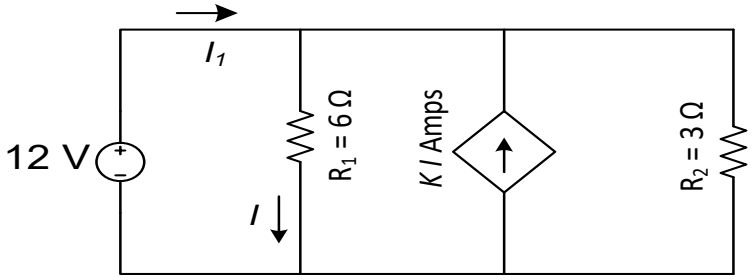


[Ans.: $V_0 = -6$ V, $I_1 = -2$ A, $I_2 = -4$ A, $P_{6\Omega} = 24$ W]

Q4. For the circuit given below, determine the power absorbed by each element, i.e., voltage source, current source, resistor R_1 and resistor R_2, when

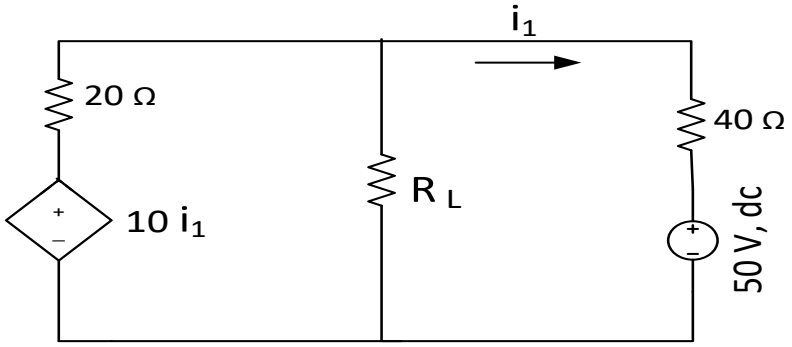
(a) K=2

(b) K= - 2



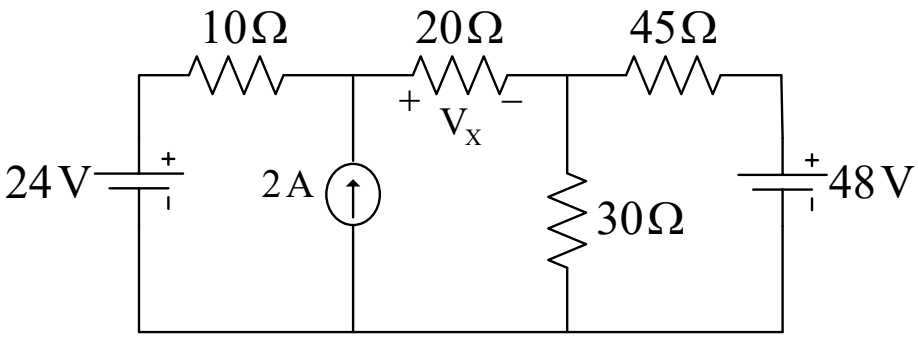
[Ans.: (a) P_{12V}= 24 W, $P_{6\Omega}$ =24 W, $P_{K\,I}$= 48 W, $P_{3\Omega}$ =48 W, (b)) P_{12V}= 120 W, $P_{6\Omega}$ =24 W, $P_{K\,I}$=- 48 W, $P_{3\Omega}$ =48 W]

Q5. With reference to the following circuit: (a) Determine the value of R_L to which a maximum power can be delivered, and (b) Calculate the voltage across R_L. Consider positive terminal at top.



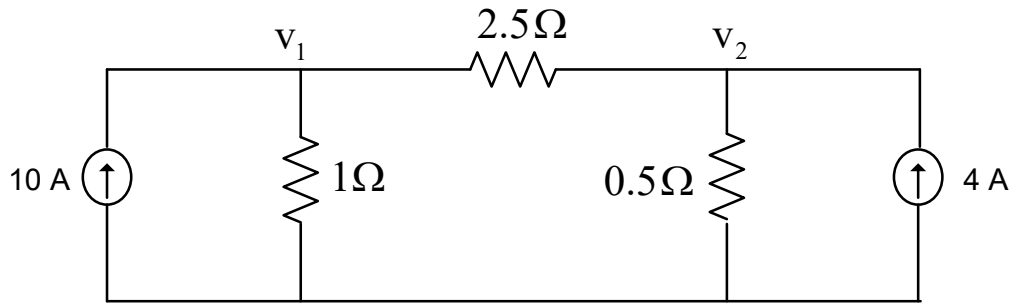
[Ans.: R_L=R_i=16 Ω, V_{RL}=5 V]

Q6. Use Superposition to find the value of Vx in the circuit of figure given below.



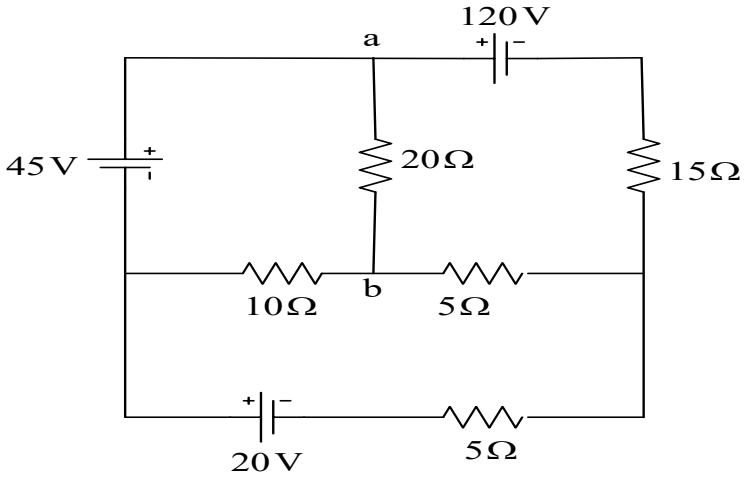
[Ans.: V_X=10.4 V]

Q7. In the circuit shown below, answer the following questions.
(a) How many meshes does this circuit have?
(b) What are the mesh currents for the inner and outer meshes?
(c) Find all mesh currents.
(d) Determine voltages V1 and V2.



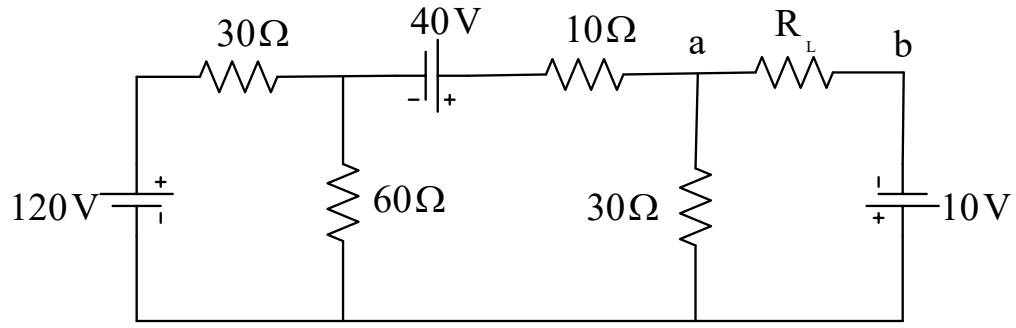
[Ans.: (a) 3, (b) 10, - 4 A (clockwise), (c) 10, 2, -4 A (clockwise) (d) 8, 3 V

Q8. For the circuit shown below, find the Thevenin equivalent circuit as seen by the 20-Ω resistor appearing between points a and b.



[Ans.: V_{Th}= 63 V, R_{th}= R_i = 4.67 Ω]

Q9. By direct application of Norton's theorem find the Norton equivalent circuit as seen by R_L in the circuit.



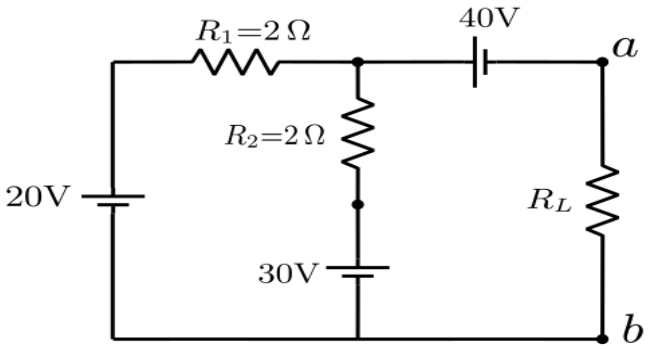
[Ans.: I_N=I_{sc}= 4.67 A, R_N=R_i= 15 Ω]

Q10. Consider the circuit shown in the figure.

(a) For R_L = 3 Ω, find the power delivered to R_L.

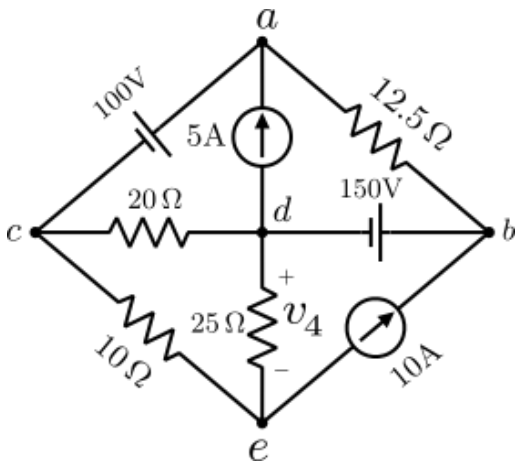
(b) Find the value of R_L for which maximum power is delivered to R_L. Also find the maximum power.

(c) Find the value of R_L for which the power delivered is P_{max} / 2 (half of the maximum power).



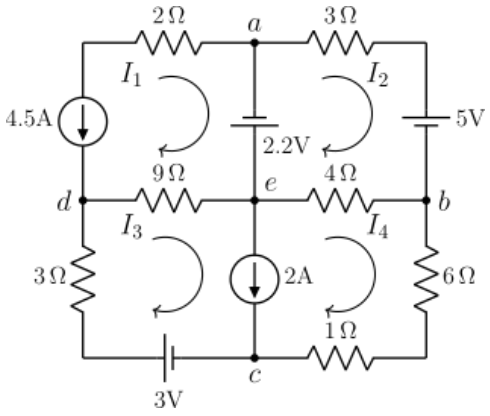
[Ans.: 42.18 W, 1 Ω, R_L=5.8 Ω, 0.172]

Q11. In the circuit given below, find currents in all branches and voltage v_4. (Hint: nodes a and c, b and d form super nodes).



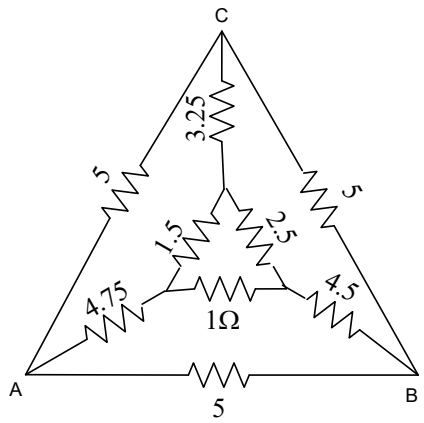
[Ans.: I_{ac} = 8.063 A, I_{ba} = 3.063 A, I_{da} = 5.000 A (given), I_{cd} = 0.586 A, I_{ce} = 7.477 A, I_{bd} = 6.937 A, I_{de} = 2.523 A, I_{eb} = 10.000 A (given), v_4 = 63.06 V]

Q12. For the network shown in the figure, use mesh analysis to determine all branch currents, then compute the power delivered by each source and prove that the total power supplied equals the total power consumed.



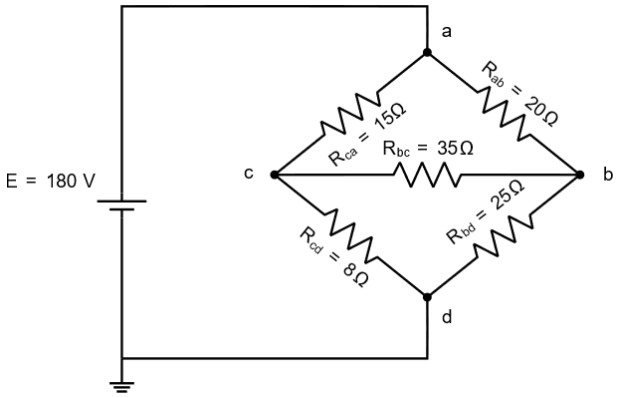
[Ans.: I_{ad} = 4.500 A (fixed by 4.5 A source), I_{ba} = 2.839 A, I_{be} = 0.329 A, I_{de} = 3.332 A, I_{cd} = 1.168 A, I_{cb} = 3.168 A, I_{ea} = 1.661 A, I_{ec} = 2.000 A (given), P(4.5A) = +185.363 W, P(2A) = +46.978 W, P(5V) = +14.193 W, P(2.2V) = −3.655 W, P(3V) = −3.503 W, P_supplied = P_consumed = 239.376 W]

Q13. In the network shown in figure below, find the resistance between the points A and B. The value of each resistor shown in figure is given in ohms.



[Ans.: R_{AB}=2.5 Ω]

Q14. For the circuit shown in the figure, find the current through each element and the current supplied by the source, using the following methods: (i) Star–Delta (Wye–Delta) transformation, (ii) Mesh analysis, and (iii) Nodal analysis. Also, prove that the power consumed by the resistive elements is equal to the power supplied by the source.



[Ans.: Iac = 7.573 A, Iab = 4.405 A, Ibc = 0.729 A, Icd = 8.301 A, Ibd = 3.676 A, Isource = 11.978 A, P(Rca) = 860.2 W, P(Rab) = 388.1 W, P(Rcb) = 18.6 W, P(Rcd) = 551.3 W, P(Rbd) = 337.9 W, P_supplied = P_consumed ≈ 2156 W]